NASA Experience and Careers

Exploring Opportunities in Human Spaceflight and Aerospace Engineering

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NASA Experience and Careers

FIRST ROBOTICS COMPETITION

Minneapolis, MN

Honeywell

College of Engineering
UNIVERSITY OF WISCONSIN–MADISON

AIAA
SHAPING THE FUTURE OF AEROSPACE



1

2

Madison, WI











NASA

3

Houston, TX

Honeywell

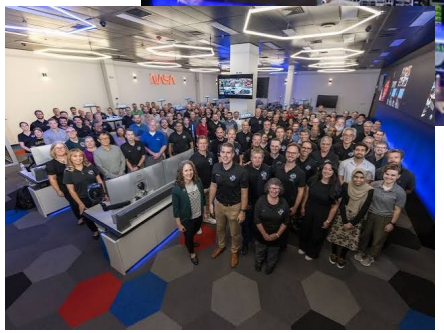
Johnson Space Center (JSC) – Houston, TX

- NASA's center for human spaceflight
- Home to Mission Control and astronaut training
- Located in Houston, Texas
- Leading development of spacecraft for Moon and Mars missions
- Over 10,000 civil service and contractor employees









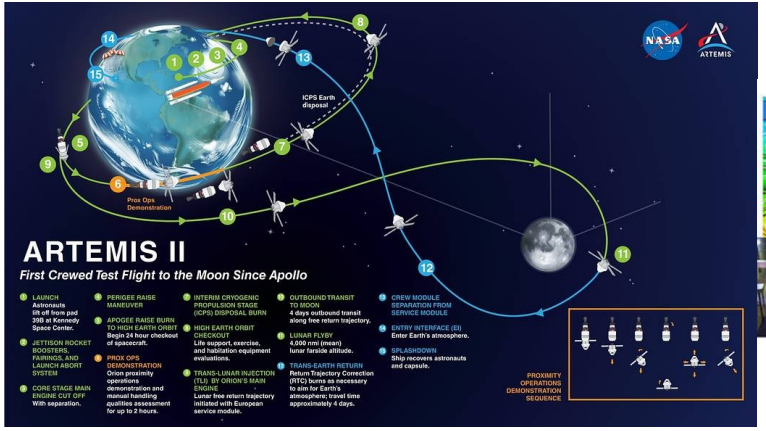


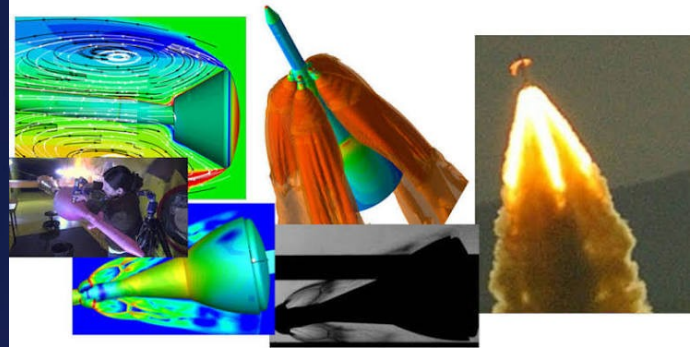
JSC EG – Aeroscience and Flight Dynamics Division

Key Technical Disciplines

- Aerodynamics, Aerothermal - spacecraft flight in atmospheres
- Guidance, Navigation & Control (GN&C) – Vehicle dynamics, control
- Flight Mechanics - Orbital dynamics and trajectory design
- Abort Systems - Crew safety systems for emergencies
- Parachute Systems - Safe landing technologies
- Independent Verification & Validation – Independent Modeling/Sim





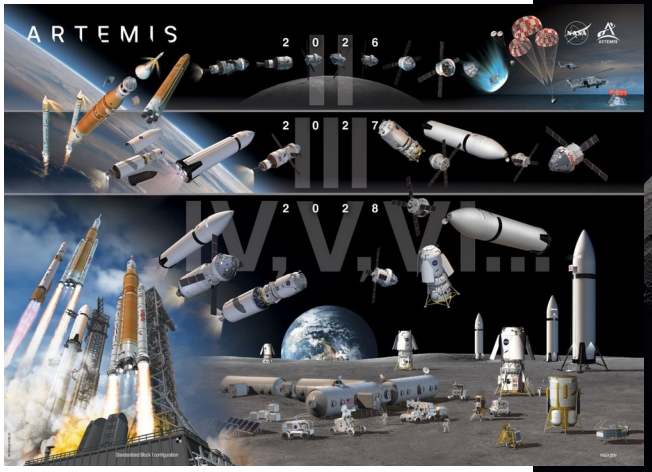




JSC EG – Aeroscience and Flight Dynamics Division

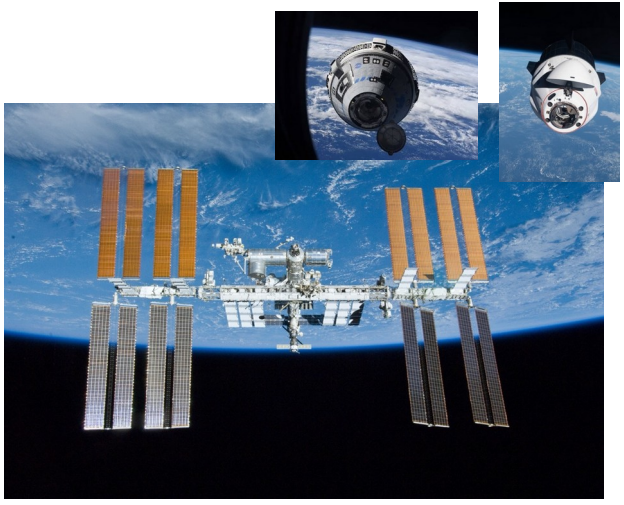
Programs We Support

- Artemis - Orion Spacecraft (returning humans to the Moon)
- International Space Station (ISS)
- Commercial Crew Program - SpaceX Dragon and Boeing Starliner
- Moon Base
- Future exploration missions to Mars and beyond





ARTEMIS II VIDEO



Project Spotlight: Morpheus Lander

- Vertical takeoff and landing test vehicle
- Tested autonomous landing technologies
- Flight campaigns at JSC and Kennedy Space Center
- Hands-on work: system design, integration, flight testing
- Pathfinder for commercial and human moon missions





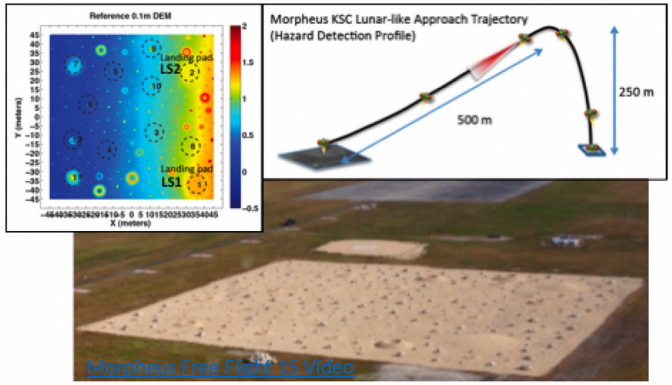


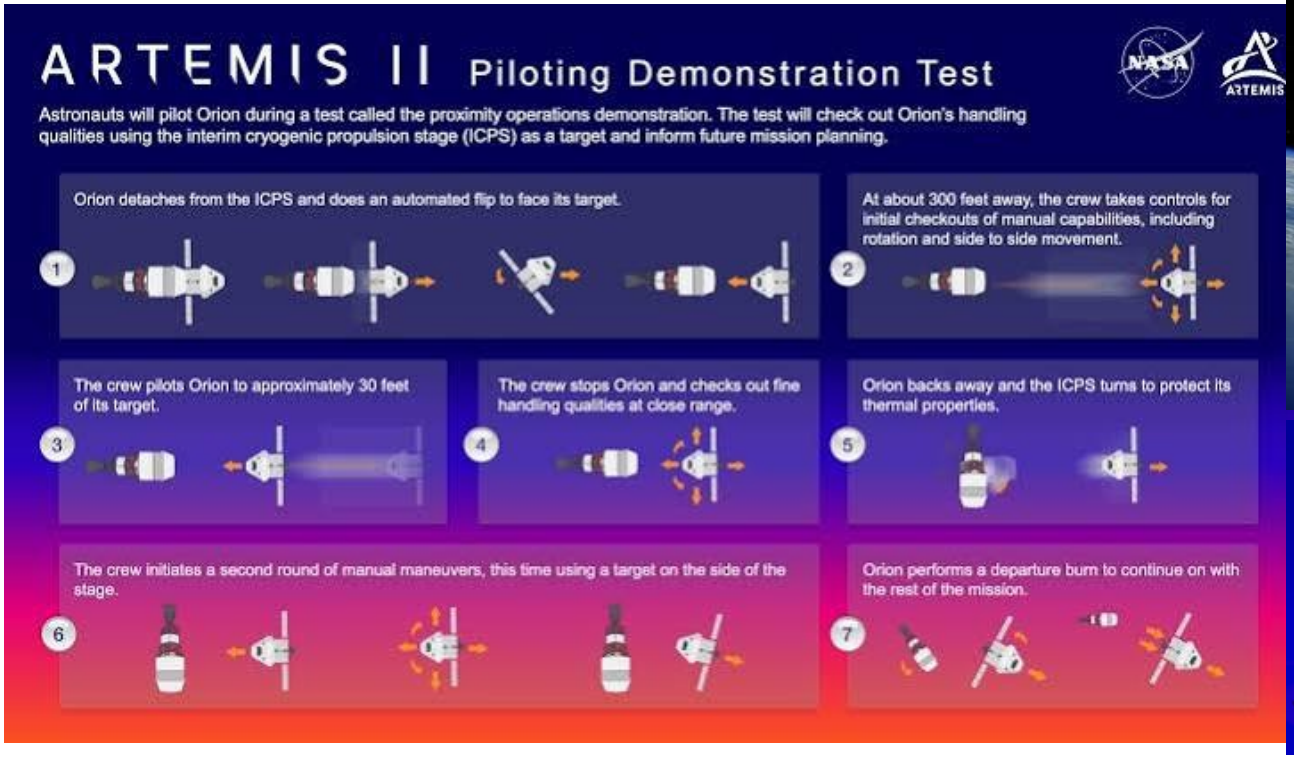


Figure 7 – HDP Trajectory and Hazard Field Overview

https://ntrs.nasa.gov/api/citations/20140003934/downloads/20140003934.pdf

Morpehus Lander Video

Project Spotlight: Artemis II Prox Ops Demo





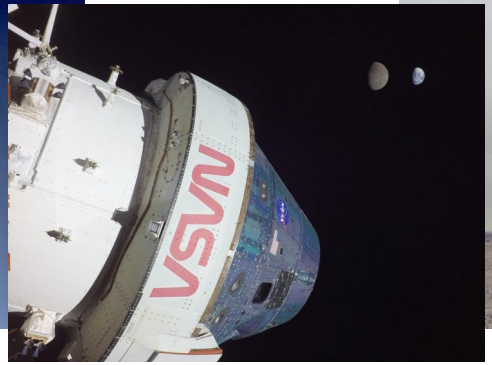


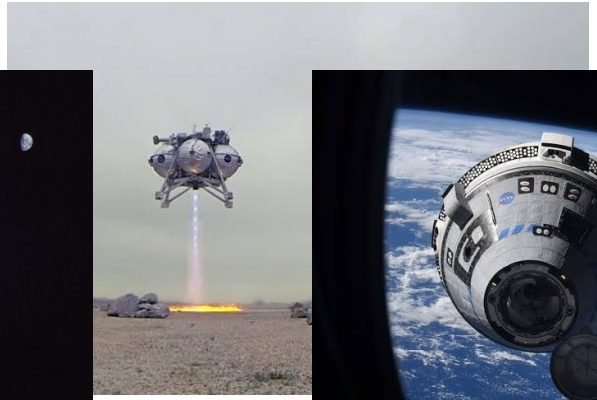
[Flight Day Video (31:45)]

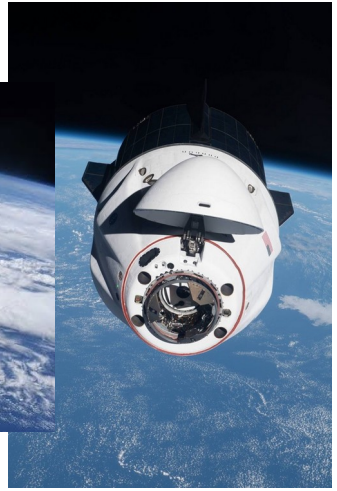
Example Career Progression at NASA

- Entry Level: Project engineer on X-38 and Shuttle programs
- Mid-Career: GN&C Subsystem Lead on Orion and Morpheus
- Senior: System Manager for Commercial Crew Program
- Leadership: Deputy Division Chief overseeing multiple programs









Getting Started: Pathways to NASA

Relevant Degree Programs:

- Aerospace Engineering
- Mechanical Engineering
- Engineering Mechanics
- Electrical Engineering
- Computer Science / Software Engineering
- Physics and Applied Mathematics

NASA Career Opportunities and Tips:

- NASA Internships (OSTEM, Pathways)
- Contractor positions - Aerospace companies supporting NASA
  - JSC Prime Contractor is Amentum
- Build relevant skills: Engineering, Programming, CAD, Systems Engineering
- Join student organizations: AIAA, University and NASA design competitions, Conference Papers

Pathways Internship

Paid internships that are direct pipelines to full-time employment at NASA upon graduation.

Pathways Internships are federal internships designed to prepare students for a career at NASA. Pathways interns rotate through multiple assignments and can be converted directly into full-time civil service roles after they successfully complete the program.

Click Here to Learn More About Pathway Internships

OSTEM Internship

Paid internships that allow college-level students to contribute to the agency's mission to advance science, technology, aeronautics, and space exploration.

OSTEM Internships offer short-term, hands-on opportunities for students to work on real NASA projects. These experiences help to build technical and professional skills while learning from NASA mentors.

Join our NASA team and gain valuable on-the-job experience, build your resume, and strengthen your career readiness. We offer three sessions annually, so visit our website often for opportunities.

Summer 2027 Application Deadline: March 1, 2027 (11:59 p.m. ET)
Fall 2027 Application Deadline: May 24, 2027 (11:59 p.m. ET)

Click Here to Explore Our Opportunities and Apply

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The Future of Space Exploration

- Artemis missions are returning humans to the Moon
- Establishing a sustainable lunar presence
- Preparing for human missions to Mars
- Your generation will make this happen!

Questions?